Hartogs' lemma: for every set A there is an ordinal not injecting into A

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/olp-set-theory-choice/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Card cut from the FREE slice of the lit pool for unit olp-set-theory-choice; cross-chapter references resolved to their pinned source units.

Editorial scope. Single lemma card cut from the Open Logic Project unit olp-set-theory-choice (pinned-original-source.tex); statement and proof are the author's own text at source lines 674-676 and 678-707.

Allowed standard prerequisites (editorial).

- Rank computation for subsets and products (set-theory-ord-arithmetic chapter, Lemma rankcomputation, source line 820). cited at proof lines 679-681. Gives the rank bound placing every pair (B,R) with B a subset of A and R a subset of the square of B inside the stage V of rank(A)+4.
- Ordinal representation theorem (set-theory-ordinals chapter, Theorem thmOrdinalRepresentation, source line 1020). cited at proof lines 687-688. Guarantees each well-ordering (B,R) has an order type, so Replacement can collect the order types of C into a set.
- Corollary: a transitive set of ordinals is an ordinal (set-theory-ordinals chapter, Corollary corordtransitiveord, source line 534). cited at proof line 692. Identifies the collected set alpha as an ordinal once transitivity is checked.
- Initial segments of well-orderings (set-theory-ordinals chapter, Lemma wellordinitialsegment, source line 799). cited at proof lines 695-696. Shows predecessors gamma of an order type are themselves order types of initial segments, which is what makes alpha transitive.

Source reference. Original lemma 0.15: sources/olp-set-theory-ord-arithmetic/pinned-original-source.tex, source lines 820–820. Cited at proof lines 679-681: supplies the rank bound V of rank(A)+4 containing every candidate pair (B,R).

Source reference. Original theorem 0.26: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 1020–1020. Cited at proof lines 687-688: guarantees order types exist so Replacement can collect them.

Source reference. Original corollary 0.6: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 534–534. Cited at proof line 692: a transitive set of ordinals is an ordinal.

Source reference. Original lemma 0.17: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 799–799. Cited at proof lines 695-696: predecessors of an order type are order types of initial segments, giving transitivity of alpha.

Lemma 0.1 (*in* ZF)**.** *For any set A, there is an ordinal α such that $\alpha \not\preceq A$*

Proof. If $B \subseteq A$ and $R \subseteq B^2$, then $\langle B, R\rangle \subseteq V_{\operatorname{rank} A+4}$ by 0.15 . So, using Separation, consider:

$$C = \{\, \langle B, R\rangle \in V_{\operatorname{rank} A+5} \mid B \subseteq A \text{ and } \langle B, R\rangle \text{ is a well-ordering} \,\}$$

Using Replacement and 0.26 , form the set:

$$\alpha = \{\, \operatorname{ord}(B, R) \mid \langle B, R\rangle \in C \,\}.$$

By 0.6 , α is an ordinal, since it is a transitive set of ordinals. After all, if $\gamma \in \beta \in \alpha$, then $\beta = \operatorname{ord}(B, R)$ for some $B \subseteq R$, whereupon $\gamma = \operatorname{ord}(B_b, R_b)$ for some $b \in B$ by 0.17 , so that $\gamma \in \alpha$.

For reductio, suppose there is injection $f\colon \alpha \to A$. Then, where:

$$\begin{aligned} B &= \operatorname{ran} f \\ R &= \{\, \langle f(\alpha), f(\beta)\rangle \in A \times A \mid \alpha \in \beta \,\}. \end{aligned}$$

Clearly $\alpha = \operatorname{ord}(B, R)$ and $\langle B, R\rangle \in C$. So $\alpha \in \alpha$, which is a contradiction. □